

CAPE GRIM, Australia

WMO: 959540

Lat: 40.6792S

Lon: 144.6903E

Elev: 87

StdP: 100.28

Time Zone: 10.00 (AUS)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.5	6.4	0.8	4.0	7.4	1.8	4.3	7.9	21.2	9.3	19.2	10.2	6.4	110	0.895

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.9	22.2	18.5	21.1	17.6	20.1	17.1	19.1	20.9	18.4	20.2	17.8	19.5	10.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.4	19.7	17.7	12.8	19.3	17.0	12.3	18.9	54.7	20.9	52.5	20.3	50.6	19.5	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	16.6	15.0	3.5	25.3	0.9	2.1	2.9	26.8	2.3	28.0	1.8	29.2	1.2	30.7		
(5)			2.2	20.8	0.9	1.0	1.6	21.5	1.1	22.0	0.7	22.6	0.0	23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	16.2	16.5	15.9	14.2	12.3	11.0	10.5	10.1	10.8	11.6	13.0	14.7	(6)
(7)		DBStd	2.90	2.01	1.98	2.23	1.85	1.71	1.63	1.44	1.61	1.59	1.67	2.07	2.02	(7)
(8)		HDD10.0	55	0	0	0	0	1	10	12	19	9	3	1	0	(8)
(9)		HDD18.3	1944	72	58	79	124	186	221	243	255	226	208	159	114	(9)
(10)		CDD10.0	1172	193	181	184	127	74	38	27	22	33	54	92	146	(10)
(11)		CDD18.3	21	7	6	6	1	0	0	0	0	0	0	1	2	(11)
(12)		CDH23.3	17	10	3	3	0	0	0	0	0	0	0	0	1	(12)
(13)		CDH26.7	1	1	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	8.6	8.5	8.8	8.5	7.8	8.5	8.1	9.4	8.8	8.4	8.6	9.0	8.6	(14)
(15)	Precipitation	PrecAvg	945	42	41	56	77	93	105	123	113	94	79	65	53	(15)
(16)		PrecMax	1297	112	159	153	180	255	177	224	236	221	150	150	145	(16)
(17)		PrecMin	664	12	4	7	25	28	39	48	28	22	30	20	11	(17)
(18)		PrecStd	152	22	35	36	41	51	33	46	44	40	33	29	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	23.4	23.4	20.0	17.1	14.8	14.6	14.5	16.0	18.1	20.7	22.4	(19)
(20)			MCWB	19.4	19.2	19.4	16.2	14.2	13.1	12.7	12.2	13.1	14.1	16.9	18.0	(20)
(21)		2%	DB	22.0	21.7	21.5	18.7	16.2	14.2	13.7	13.6	15.0	16.5	19.0	20.5	(21)
(22)			MCWB	18.6	18.1	18.1	16.1	13.9	12.6	11.9	11.7	12.6	13.6	16.5	16.9	(22)
(23)		5%	DB	20.7	20.7	20.3	17.7	15.6	13.7	13.2	13.1	14.1	15.2	17.4	18.9	(23)
(24)			MCWB	17.5	17.5	17.6	15.5	13.6	12.2	11.4	11.2	12.2	13.3	15.4	16.1	(24)
(25)		10%	DB	19.4	19.7	19.3	17.0	15.0	13.3	12.7	12.5	13.2	14.3	16.2	17.9	(25)
(26)			MCWB	16.7	17.0	17.1	15.3	13.3	11.7	10.9	10.8	11.7	12.6	14.6	15.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	19.7	19.8	17.5	15.5	14.1	13.3	12.7	13.7	15.1	18.1	18.5	(27)
(28)			MCDWB	22.6	22.4	22.5	18.6	16.1	14.3	14.1	13.8	14.8	16.3	19.3	21.0	(28)
(29)		2%	WB	18.7	18.9	19.0	16.7	14.6	13.1	12.4	12.1	13.0	14.3	16.7	17.7	(29)
(30)			MCDWB	20.8	20.6	20.6	17.7	15.4	13.6	13.2	13.2	14.3	15.6	18.4	19.5	(30)
(31)		5%	WB	18.0	18.2	18.1	16.2	14.0	12.5	11.8	11.6	12.5	13.5	15.7	16.9	(31)
(32)			MCDWB	20.3	19.9	19.7	17.3	15.1	13.3	12.6	12.7	13.7	14.9	17.2	18.5	(32)
(33)		10%	WB	17.2	17.6	17.3	15.5	13.5	12.0	11.2	11.1	11.9	12.8	14.5	15.9	(33)
(34)			MCDWB	19.1	19.3	18.8	16.8	14.7	13.0	12.3	12.2	13.0	14.2	16.2	17.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.7	4.9	4.7	4.0	3.4	3.3	3.2	3.6	4.2	4.0	4.6	(35)	
(36)			MCDBR	6.8	6.1	6.1	5.1	3.6	3.2	3.1	3.7	4.3	5.5	5.8	6.4	(36)
(37)			MCWBR	4.3	4.1	4.0	3.8	3.4	3.1	2.9	3.3	3.6	4.1	3.8	4.3	(37)
(38)		5% WB	MCDWB	6.0	5.1	5.3	4.4	3.6	2.9	2.9	3.4	3.8	5.3	5.4	5.7	(38)
(39)			MCWBR	4.2	4.3	3.8	3.6	3.4	3.1	3.2	3.4	3.5	4.2	3.6	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.347	0.339	0.332	0.326	0.313	0.306	0.309	0.322	0.359	0.339	0.340	0.340	(40)	
(41)		taud	2.528	2.568	2.586	2.578	2.597	2.602	2.598	2.560	2.427	2.512	2.529	2.543	(41)	
(42)		Ebn at Noon	982	964	929	865	811	784	808	856	881	954	982	994	(42)	
(43)		Edh at Noon	109	100	90	78	65	60	64	78	103	104	108	108	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.71	5.85	4.39	3.00	1.93	1.53	1.67	2.49	3.66	5.15	6.15	6.74	(44)	
(45)		RadStd	0.42	0.28	0.21	0.17	0.11	0.11	0.10	0.19	0.20	0.28	0.36	0.44	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		+0.29	N/A	+0.42	+0.59	+0.49	+0.51	N/A	-81	+92	+8

Nomenclature: See separate page